MILITARY SPECIFICATION SHEET

FILTER ELEMENT, FLUID PRESSURE, HYDRAULIC, AIRCRAFT, DISPOSABLE, 5 MICRON ABSOLUTE

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-83861.

REQUIREMENTS:

Bubble Point. The initial and burst out bubble point values shall meet the values established by the activity responsible for qualification and shall be determined as specified in 4.6.1 of general specification.

Filtration. The filter elements shall remove 100 percent of all contaminants over 5 microns in size and 95 percent by weight of add contaminats.

Cold Start. The filter element shall withstand 10 flow cycles as specified in 4.6.6 of general specification at -65°F.

Element Pressure Drop. The pressure drop through the clean element using MIL-H-5606 hydraulic fluid at a temperature of 100°F ± 10°F shall not exceed 10.0 psid at rated flow.

Element Dirt Capacity. The filter element shall meet the following dirt capacity schedule:

Element Part Number	Contaminant - Add Increments A-D Dust (grams)	Minimum Weight of Test Dust (grams)
M83861/1	0.4	3.0 grams to 40 psid

FSC 1650

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Element Collapse. The filter element shall withstand a differential pressure of 150 psid.

Description. The filter element shall be as shown in Figure 1 and as specified herein.

Flow Fatigue. The filter element shall withstand the flow fatigue cycles specified at 300 cpm and at rated flow (± 33%) at each differential pressure as follows:

Cycles	Pressure Drop (psid)
1,125,000	20
112,000	50
12,500	100
500	150

Media Migration. There shall be no media migration traceable to the filter element when tested at vibration frequencies specified in 4.6.9 of general specification.

Maximum Particle Passed. The maximum particle passed shall be no greater than 5 microns.

TABLE I. CONTAMINANTS

Flow Rate (gpm)	Degree of Filtration Test		Maximum Particle Test	Pressure Buildup Test	
	Add Rate F-9 Beads (gms)	Add Rate AC Dust (gms)	Add Rate Carbonyl Iron E (gms)	Add Rate (gms)	Min AC Dust to 40 psid
22	0.15	0.15	0.30	0.4	3.0

Part Number. The part number consists of an "M" plus the basic number of this specification sheet.

Part Number: M83861/1

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 99

Project Number:
1650-F063

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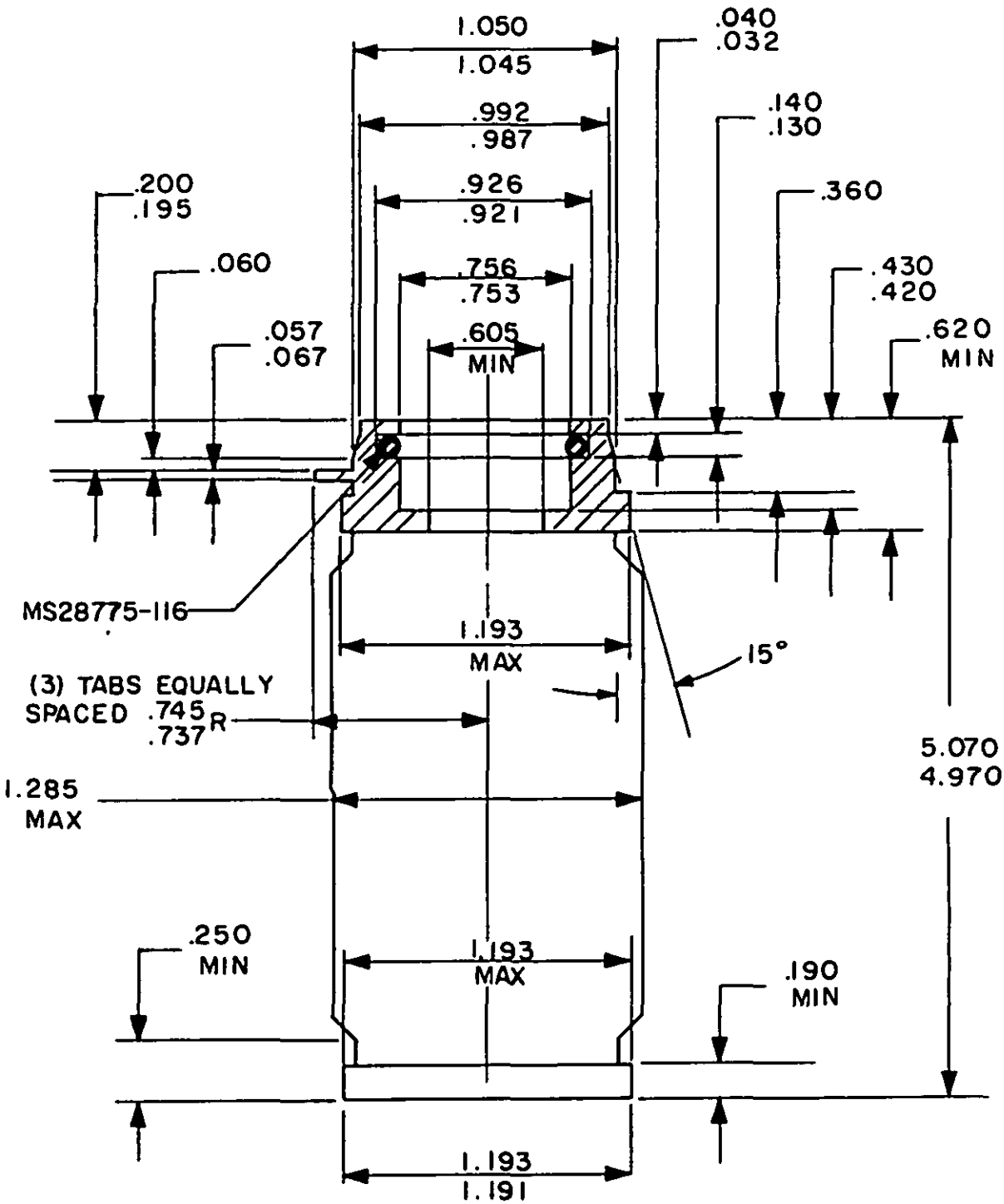


FIGURE 1.

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